

Causal Rigidity and the Single-Unit Universe: Integrating the Alexandrov-Zeeman and Unruh Clock Scales

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We unify two complementary viewpoints on relativistic spacetime and the counting of fundamental constants. Operationally, Matsas, Pleitez, Saa, and Vanzella (MPSV) have recently argued that relativistic spacetime requires only a single fundamental dimensional constant. Mathematically, theorems due to Alexandrov and Zeeman demonstrate that the light-cone structure determines the spacetime geometry only up to a conformal factor. We show that these approaches are mutually reinforcing: the Alexandrov-Zeeman theorems establish the rigid conformal structure of spacetime, while the “bona fide clock” required by MPSV serves the necessary mathematical role of breaking the dilation symmetry. We provide a formal derivation proving that the normalization of a single clock worldline is sufficient to select a unique metric from the conformal class, thereby clarifying that the number of fundamental constants is exactly one.

I. INTRODUCTION

The question “How many fundamental dimensional constants does physics require?” has been a subject of enduring debate, most famously in the “trialogue” between Duff, Okun, and Veneziano (DOV) [1]. While there is consensus that only dimensionless quantities are ultimately observable, disagreements persist regarding the number of independent *standards* (or base units) necessary to describe dimensionful observables.

Recently, Matsas, Pleitez, Saa, and Vanzella (MPSV) [2] revisited this issue from a *spacetime-based* perspective. They argue that once a relativistic spacetime is assumed as the starting point, the units required to construct that spacetime are sufficient to express all observables. Specifically, they conclude that in a relativistic spacetime, the number of fundamental dimensional constants is *one*.

Parallel to this metrological debate, the mathematical relativity community has long established—via theorems by Alexandrov [3] and Zeeman [4]—that the null (light-cone) structure of spacetime is incredibly rigid. These theorems state that any map preserving the causal order is a Lorentz transformation up to a global dilation (scale factor).

In this paper, we unify these two perspectives. In Sec. II, we review the operational argument provided by MPSV. In Sec. III, we review the geometric constraints imposed by causal automorphisms. Finally, in Sec. IV, we provide a formal derivation demonstrating that the “bona fide clock” of MPSV is the precise mathematical object required to break the conformal symmetry left open by incidence geometry.

II. THE OPERATIONAL PERSPECTIVE: MEASURING SPACE WITH TIME

MPSV argue that to construct a relativistic spacetime physically, one requires specific apparatuses. Unlike Galilean spacetime, which requires both rulers and clocks (because space and time are decoupled), relativistic spacetime requires only “bona fide clocks.”

To prove that spatial rulers are redundant, MPSV utilize a protocol attributed to William George Unruh. Consider three inertial clocks C_1, C_2, C_3 :

1. C_1 travels from an observer A to B .
2. C_2 travels from B back to A .
3. C_3 remains at rest with A .

If τ_1 and τ_2 are the proper time durations measured by the traveling clocks, and τ_3 is the duration measured by the stationary clock, the spatial distance D between A and B is given by the height corresponding to the base τ_3 of the triangle formed by sides τ_1, τ_2, τ_3 (derived from Heron’s formula [2]):

$$D = \frac{\sqrt{(\tau_3^2 - \tau_1^2 - \tau_2^2)^2 - 4\tau_1^2\tau_2^2}}{2\tau_3}. \quad (1)$$

This formula yields a spatial length in units of time (seconds). It demonstrates physically that the spatial metric can be derived entirely from time measurements, rendering the meter a “disposable” unit and c a mere conversion factor. When the two traveling clocks record the same proper time, $\tau_1 = \tau_2 \equiv \tau$, the expression simplifies to $D = (\tau_3/2)\sqrt{1 - (2\tau/\tau_3)^2}$.

III. THE GEOMETRIC PERSPECTIVE: ALEXANDROV-ZEEMAN THEOREMS

We now turn to the mathematical structure of Minkowski spacetime. We consider the manifold $M \simeq$

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$\mathbb{R}^n$ (with $n \geq 3$), equipped with the standard flat metric $\eta_{ab} = \text{diag}(-1, 1, \dots, 1)$. Throughout this paper, we adopt the “mostly plus” signature convention $(-, +, +, \dots, +)$, so that timelike vectors have negative squared norm and spacelike vectors have positive squared norm. The defining feature of relativistic geometry is the causal structure, encoded either by the incidence of light cones or the partial causal ordering of events.

A fundamental question in mathematical relativity is the extent to which this causal structure alone determines the geometry. This is resolved by a class of rigidity results known as Alexandrov-Zeeman type theorems. We state these results with their precise hypotheses, as the conditions are essential to their validity.

A. Alexandrov: The Rigidity of Light Cones

In the 1950s, A. D. Alexandrov proved that the geometry of spacetime is encoded in the “skeleton” of light rays [3].

Theorem 1 (Alexandrov). *Let $f : M \rightarrow M$ be a bijection of the entire spacetime manifold $M \simeq \mathbb{R}^n$ with $n \geq 3$. If f maps null lines to null lines (i.e., preserves the light-cone incidence structure), then f is a linear conformal automorphism.*

The requirement that f be a *global* bijection of the entire spacetime is crucial. Local maps preserving null structure need not extend to conformal transformations; indeed, general relativity admits local diffeomorphisms that preserve the causal structure within a region but are not globally conformal. The theorem’s power derives precisely from its global character.

B. Zeeman: Causality Implies the Lorentz Group

In 1964, E. C. Zeeman strengthened this perspective by focusing on the partial ordering of events [4]. We define the causal relations as follows: $x \prec y$ (read “ x chronologically precedes y ”) if a future-directed timelike or null signal can travel from x to y , and $x \succ y$ if $y \prec x$.

Theorem 2 (Zeeman). *Let $f : M \rightarrow M$ be a bijection of the entire Minkowski spacetime $M \simeq \mathbb{R}^n$ with $n \geq 3$. Suppose f preserves the full causal order, meaning:*

$$x \prec y \iff f(x) \prec f(y) \quad \text{and} \quad x \succ y \iff f(x) \succ f(y). \quad (2)$$

Then f is a composition of:

1. an orthochronous Lorentz transformation $\Lambda \in O^\uparrow(1, n-1)$,
2. a spacetime translation $a \in \mathbb{R}^n$, and
3. a global dilation $x \mapsto \lambda x$ with $\lambda > 0$.

Several remarks on the hypotheses are in order:

- *Dimensionality:* The restriction $n \geq 3$ is essential. In $(1+1)$ -dimensional spacetime, the theorem fails: the null lines form two independent families, and maps can independently rescale each family while preserving causal order.
- *Bijectivity and global domain:* The map must be defined on and surject onto the *entire* Minkowski space. This excludes local causal isomorphisms, which are far more numerous and less constrained.
- *Full causal structure:* Preserving only the future-directed relation $\prec$ is insufficient. One must preserve both temporal orientations; equivalently, the map must preserve the relation of being causally connectable (whether to the past or future). Maps preserving only one direction could, in principle, include time-reversing transformations not generated by the orthochronous Lorentz group.

C. Limitations and Extensions

It is important to recognize the scope of these theorems. They apply to flat Minkowski spacetime as a global entity. Extensions to curved spacetimes require substantial additional machinery: In general relativity, the celebrated result of Malament [5] establishes that the causal structure of a distinguishing spacetime determines the metric up to a conformal factor, but the proof strategy and required conditions differ significantly from the flat case. For the present discussion, we restrict attention to special relativity, where the Alexandrov-Zeeman framework applies directly.

D. The Conformal Ambiguity

Both theorems lead to the same physical conclusion: the causal structure determines the metric g_{ab} up to a *conformal factor*. Under the strict conditions of mapping the whole of Minkowski space to itself bijectively, this factor reduces to a global positive constant λ :

$$g'_{ab} = \lambda^2 \eta_{ab}. \quad (3)$$

This leads to a profound physical consequence: A universe defined solely by light rays (or causal order) cannot distinguish a second from a millennium. The transformation $x^\mu \rightarrow \lambda x^\mu$ for any $\lambda > 0$ is a valid symmetry of the causal structure. The “shape” of physics—the pattern of which events can influence which—is preserved, but all durations and lengths are uniformly rescaled. To distinguish absolute time scales, one must introduce additional structure that breaks this dilation symmetry.

IV. FORMAL UNIFICATION: THE CLOCK AS SYMMETRY BREAKER

We now formalize the connection between the MPSV clock and the Alexandrov-Zeeman conformal factor. We show that specifying the normalization of a single clock worldline is sufficient to fix the global scale λ .

A. Definitions

Definition 1 (Conformal Class). *Let η be the standard Minkowski metric. The conformal class $[\eta]$ determined by the causal structure is the set of metrics $\{\lambda^2\eta : \lambda > 0\}$.*

Definition 2 (Proper Time). *Let $\gamma : I \rightarrow M$ be a smooth timelike curve parameterized by an arbitrary parameter $s \in I \subseteq \mathbb{R}$. The proper time along γ with respect to a metric g is:*

$$\tau_g(s) = \int_{s_0}^s \sqrt{-g(\dot{\gamma}(s'), \dot{\gamma}(s'))} ds', \quad (4)$$

where $s_0 \in I$ is a reference parameter value (corresponding to the initial event) and dot denotes d/ds' .

Definition 3 (Bona Fide Clock). *A bona fide clock consists of:*

1. *a timelike worldline $\Gamma \subset M$, considered as a point set;*
2. *a physical mechanism that produces readings $\tau_{\text{clock}} : \Gamma \rightarrow \mathbb{R}$ along the worldline.*

The clock is ideal with respect to the physical metric $g_\star$ if its readings coincide with the proper time:

$$\tau_{\text{clock}}(\gamma(s)) = \tau_{g_\star}(s) + \text{const}, \quad (5)$$

for any parameterization $\gamma : I \rightarrow \Gamma$.

The key distinction is that the worldline Γ and the reading function τ_{clock} are *physically given* by the apparatus, independent of any coordinate choice or metric assumption. The condition (5) then becomes a constraint that the unknown metric $g_\star$ must satisfy.

B. Uniqueness Theorem

Theorem 3 (Single Clock Fixes Scale). *Let $M \cong \mathbb{R}^4$ be equipped with a causal order determining the conformal class $[\eta]$. Let $(\Gamma, \tau_{\text{clock}})$ be a single ideal clock following an inertial trajectory. Then the physical metric $g_\star \in [\eta]$ is uniquely determined.*

Proof. Since the causal structure fixes the metric up to a global scale (via the Alexandrov-Zeeman theorems for $n \geq 3$), the physical metric must take the form:

$$g_\star = \lambda^2 \eta, \quad \lambda > 0, \quad (6)$$

where η is an arbitrary reference metric in standard Minkowski coordinates.

Let $\gamma : \mathbb{R} \rightarrow \Gamma$ be *any* smooth parameterization of the worldline by a parameter s . The clock's physical mechanism provides the reading τ_{clock} along the worldline; differentiating with respect to s yields the empirically determined tick-rate $d\tau_{\text{clock}}/ds$. For an inertial clock, this rate is constant.

Differentiating the ideal clock condition (5) with respect to s yields:

$$\frac{d\tau_{\text{clock}}}{ds} = \sqrt{-g_\star(\dot{\gamma}, \dot{\gamma})} = \sqrt{-\lambda^2 \eta(\dot{\gamma}, \dot{\gamma})} = \lambda \sqrt{-\eta(\dot{\gamma}, \dot{\gamma})}. \quad (7)$$

Define $\alpha(s) \equiv \sqrt{-\eta(\dot{\gamma}(s), \dot{\gamma}(s))}$, which is the norm of the tangent vector in the reference metric. This quantity depends on the arbitrary parameterization s , but the ratio

$$\lambda = \frac{d\tau_{\text{clock}}/ds}{\alpha(s)} \quad (8)$$

is *independent* of the choice of parameterization. To see this, consider a reparameterization $s \rightarrow \tilde{s}(s)$. Then $d\tau_{\text{clock}}/d\tilde{s} = (d\tau_{\text{clock}}/ds)(ds/d\tilde{s})$ and $\tilde{\alpha} = \alpha \cdot |ds/d\tilde{s}|$, so their ratio is invariant.

Since $d\tau_{\text{clock}}/ds$ is physically given by the clock mechanism and $\alpha(s)$ is computed from the reference coordinates, the scale factor λ is uniquely determined. Thus $g_\star = \lambda^2 \eta$ is unique. $\square$

C. The Hierarchy of Metrics

It is instructive to explicitly distinguish the roles of the metric tensors discussed above:

1. **The Reference Metric (η_{ab}):** An arbitrary mathematical auxiliary (the standard Minkowski metric) that correctly describes the causal incidence (light cones) but possesses no physical scale.
2. **The Conformal Class ($[\eta]$):** The set of all possible metrics allowed by the Alexandrov-Zeeman theorems. This is a one-parameter family:

$$g'_{ab} = \lambda^2 \eta_{ab}, \quad \forall \lambda \in (0, \infty). \quad (9)$$

At this level, the theory cannot distinguish between a second and a millennium; any value of λ preserves the causal order.

3. **The Physical Metric ($g_\star$):** The unique element of the conformal class selected by the physical apparatus. It is obtained by evaluating at the specific scale factor determined by the clock:

$$g_\star = \lambda_{\text{phys}}^2 \eta_{ab}. \quad (10)$$

Thus, while the causal structure generates the conformal equivalence class, the single bona fide clock collapses this class to the unique physical metric.

V. DISCUSSION

The formal derivation above confirms the operational intuition of MPSV. The “one fundamental constant” is the physical realization of the mathematical mechanism required to break the dilation symmetry of incidence geometry.

- *Incidence Relations (Alexandrov/Zeeman)*: These fix the conformal class $[\eta]$. Physically, they define the “shape” of spacetime and the structural role of c , but leave the scale undetermined.
- *The Clock*: This provides the constraint that the metric must reproduce the clock’s physical readings as proper time, thereby eliminating the dilation sector.

Therefore, the counting of fundamental constants in Special Relativity is exactly one because the symmetry group of the underlying structure is the Poincaré group extended by dilations. The causal structure provides the Poincaré sector, and the single unit (the clock) fixes the scale.

Finally, regarding the role of c : kinematically, MPSV [2] correctly identify c as a mere unit converter between the fundamental time and the derivative space. Yet, dynamically, c retains a privileged status. It is the unique velocity parameter that ensures the form invariance of the equations of motion, such as those of electrodynamics [6, 7]. Thus, while the clock fixes the scale of the universe, c defines the rigid phenomenal structure necessary for the clock to operationally tick in such a way as to guarantee the form invariance of the physical laws under Lorentz boosts.

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